Project: Worked examples
Title: Pipe expansion, loops and anchor forces (screening)
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1. Scope and method

Calculates the thermal growth of a pipe run between anchors, the leg or loop that absorbs it, and the force and moment on the anchors. It is a screening calculation, not a piping stress analysis.

WHAT IT CALCULATES	RELATION
Pipe section	$d_i = OD - 2 \cdot t$; $A_m = \pi/4 \cdot (OD^2 - d_i^2)$; $I = \pi/64 \cdot (OD^4 - d_i^4)$; $Z = I/(OD/2)$
Temperature change and growth	$\Delta T = T_{operate} - T_{install}$, or as entered; $L_{tot} = L$, or $L +$ L2 for a Z-bend; $\Delta L = \alpha \cdot L_{tot} \cdot \lvert \Delta T \rvert$, its sign giving the direction
Fully restrained pipe	$\sigma_r = E \cdot \alpha \cdot \lvert \Delta T \rvert$ (E at the installation temperature); $F_r = E_r \cdot \alpha \cdot \lvert \Delta T \rvert \cdot A_m$ ($E_r = E_{hot}$, else E)
Guided cantilever, required leg	$\Delta = \Delta L$ for an offset leg or Z-bend and $\Delta L/2$ for a U-loop; $L_{req} = \sqrt{(3 \cdot E \cdot OD \cdot \Delta / Sa)}$; PASS when $L_{req} \le$ the available leg H
Force and moment of the available leg	$F_{leg} = 12 \cdot E_r \cdot I \cdot \Delta / H^3$ (along the run); $M_{leg} = 6 \cdot E_r \cdot I \cdot \Delta / H^2$ (= $F_{leg} \cdot H/2$); $\sigma_{leg} = 3 \cdot E \cdot OD \cdot \Delta / H^2 = Sa \cdot (L_{req}/H)^2$
Runs of an offset leg or Z-bend	the runs bend for the leg's growth $\Delta_{leg} = \alpha \cdot \lvert \Delta T \rvert \cdot H$ under one force $F_{run} = 12 \cdot E_r \cdot I \cdot \Delta_{leg} / (L^3 + L2^3)$ along the leg (L2 = 0 for an L-bend); run i takes $L_i^3/(L^3 + L2^3)$ of Δ_{leg}, with $\sigma_i = 3 \cdot E \cdot OD \cdot \Delta_i / L_i^2$ and $M_i = F_{run} \cdot L_i/2$, and passes when $\sqrt{(\sigma_i/Sa)} \le 1$
Force on the anchors	$\sqrt{(F_{leg}^2 + F_{run}^2)}$ from the two perpendicular components; the moment reported is the largest root moment of the legs and runs
Screening and bellows	$OD \cdot Y/(L_{dev} - U)^2 \le K1$ with $Y = \alpha \cdot \lvert \Delta T \rvert \cdot U$ (offset leg or Z-bend $L_{dev} = L_{tot} + H$, $U = \sqrt{(L_{tot}^2 + H^2)}$; U-loop $L_{dev} = L + 2 \cdot H$, $U = L$); axial bellows anchor force = $P \cdot A_{eff} + k \cdot \Delta L$, with no leg check and no screening

Basis: Process piping — flexibility analysis, exemption from formal analysis[1](cl. 319.4.1 (screening relation; number to be checked against the owner's copy)); Process piping — displacement stress range and reactions[1](cl. 319.3.2 and 319.5 (reference and hot moduli; numbers to be checked against the owner's copy)); Metallic industrial piping — flexibility analysis[2](cl. 12 (number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layout	What absorbs the movement	axial bellows (no leg or loop: the bellows absorbs the movement)		drawing: DRG-M-212 rev B	any
$temp_{mode}$	Temperature change from	install and operating temperatures		assumption: Install and operating temperatures are both known.	any
OD	Pipe outside diameter	54	mm	drawing: DRG-M-212 rev B	10 … 1500 mm
t_{wall}	Pipe wall thickness	1.5	mm	drawing: DRG-M-212 rev B	0.5 … 60 mm
L	Anchor spacing: the straight length that grows (an offset leg or Z-bend: the first run, which also bends for the leg's growth)	24	m	drawing: DRG-M-212 rev B	1 … 500 m
E	Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)	120	GPa	assumption: W3 seed 'Copper (generic)', indicative; replace with the project's material record.	0.5 … 500 GPa
α	Mean coefficient of thermal expansion between the install and operating temperatures	17	um/(m·K)	assumption: W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.	1 … 250 um/(m·K)
$T_{install}$	Temperature at installation (when the run is anchored)	10	°C	assumption: Mean site temperature when the riser is anchored.	-50 … 50 °C
$T_{operate}$	Operating temperature (hot or cold)	70	°C	spec: SPEC-M-04 §3: heating flow temperature	-50 … 500 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P_b	Axial bellows governing pressure, gauge, at or above zero: the design or test pressure that governs the anchor sizing (a vacuum bellows is outside this sheet; compression and extension ratings can differ, enter the governing one)	600	kPa	spec: Heating circuit design pressure, gauge	10 ... 4000 kPa
A_{eff}	Axial bellows effective area (manufacturer's data)	40	cm²	assumption: Placeholder effective area; replace with the manufacturer's datasheet value.	1 ... 5000 cm²
k_b	Axial bellows spring rate (optional; blank = no spring force; manufacturer's data)	80	kN/m	assumption: Placeholder spring rate; replace with the manufacturer's datasheet value.	1 ... 5000 kN/m
dL_{rated}	Rated axial movement of the bellows (optional; blank = movement not checked; manufacturer's data)	40	mm	assumption: Placeholder rated axial movement; replace with the manufacturer's datasheet value.	1 ... 500 mm

3. Assumptions

A1. The result is a screen, not a piping stress analysis. (Stated scope.)
A2. The result tells whether a simple leg or loop is plausible and what it puts on its anchors. (Stated scope.)
A3. It does not replace a flexibility analysis of the real routing, including elbow flexibility, intensification, torsion, branches, supports and loads. (Stated scope.)
A4. A guided cantilever is straight, uniform, fixed against rotation at both ends and loaded only by the movement it absorbs. (Beam theory.)
A5. Its stiffness is $12 \cdot E \cdot I / H^3$ and its end moment is $F \cdot H / 2$. (Beam theory.)
A6. Elbows are taken as rigid and the leg as slender. (Beam theory.)
A7. The beam model is not valid for legs shorter than about ten diameters. (Beam theory.)
A8. No stress-intensification factor and no elbow flexibility are applied. (Stated limit.)
A9. The stress at the elbow root of a real bend is higher than the beam stress M/Z. (Stated limit.)
A10. A leg that passes here can still exceed Sa at the elbow, which is why the result is a screen. (Stated limit.)
A11. A U-loop is two identical legs in series, each a guided cantilever taking half the growth. (Loop model.)
A12. The cross leg of a U-loop is a rigid guide. (Loop model.)
A13. The force along the line is the same in both legs of a U-loop. (Loop model.)
A14. The moment M is the moment at the leg roots, the corners of the loop. (Loop model.)
A15. It reaches an anchor only where the leg ends at the anchor, as in an offset leg. (Loop model.)
A16. The growth of the loop's own legs, which the cross leg has to absorb, is not checked. (Loop model.)
A17. The run of an offset leg bends for the leg's growth as a second guided cantilever, checked against Sa. (Offset-leg and Z-bend model.)
A18. In a Z-bend the two runs are in series under one force. (Offset-leg and Z-bend model.)

A19. Each run takes its share of the leg's growth in proportion to L_i^3 and is checked as its own cantilever. (Offset-leg and Z-bend model.)
A20. The leg of a Z-bend takes the growth of both runs. (Offset-leg and Z-bend model.)
A21. The force on the anchors is the resultant of the two perpendicular forces along the run and along the leg. (Offset-leg and Z-bend model.)
A22. $anchor_{moment}$ is the largest root moment of the legs and runs, M = F·L/2 with each end fixed against rotation. (Stated limit.)
A23. It is not the moment on a frame anchor, which depends on the whole frame. (Stated limit.)
A24. A formal analysis is needed before an anchor is sized for moment. (Stated limit.)
A25. α is the mean coefficient of expansion between the install and operating temperatures. (Material library.)
A26. E is the modulus at the installation temperature, used for the stress check as the piping codes do. (The ASME process piping code, cl. 319.3.2 and 319.5 (numbers to be checked against the owner's copy).)
A27. E_{hot}, when given, is used only for the reactions on the anchors. (Code convention to be checked.)
A28. The material properties are taken from a library material applied at the right temperature. (Material library.)
A29. Sa and K1 are the owner's records from the owner's own copy of the piping code or the design basis. (No table from a standard is shipped.)
A30. No allowable stress, K1 or code constant ships. (No table from a standard is shipped.)
A31. The sheet applies the relation to the numbers entered, whatever code or load case they come from. (No table from a standard is shipped.)
A32. No weight, spring-hanger or cold-spring effect is included in the leg force. (Stated limit.)
A33. The anchor force adds support friction μ·w·L only when μ and the weight per metre are entered. (Stated limit; a warning says so when they are not.)
A34. The bellows anchor force is the pressure thrust $P \cdot A_{eff}$ plus the spring force $k \cdot \Delta L$ at the governing pressure. (Stated limit.)
A35. The governing pressure is the design or test pressure, whichever governs, because compression and extension ratings can differ. (Stated limit.)
A36. Guide friction and any other bellows load come from the manufacturer's data. (Stated limit.)
A37. $temp_{mode}$ (Temperature change from) is taken as install and operating temperatures. (Install and operating temperatures are both known.)
A38. E (Young's modulus of the pipe material at the installation (reference) temperature, for the stress check, in GPa (steel 210, copper 120, PE and PP about 1; not MPa)) is taken as 120 GPa. (W3 seed 'Copper (generic)', indicative; replace with the project's material record.)
A39. α (Mean coefficient of thermal expansion between the install and operating temperatures) is taken as 17 um/(m·K). (W3 seed 'Copper (generic)' (17e-6 1/K), indicative; replace with the project's material record.)
A40. $T_{install}$ (Temperature at installation (when the run is anchored)) is taken as 10 °C. (Mean site temperature when the riser is anchored.)
A41. A_{eff} (Axial bellows effective area (manufacturer's data)) is taken as 40 cm². (Placeholder effective area; replace with the manufacturer's datasheet value.)
A42. k_b (Axial bellows spring rate (optional; blank = no spring force; manufacturer's data)) is taken as 80 kN/m. (Placeholder spring rate; replace with the manufacturer's datasheet value.)
A43. dL_{rated} (Rated axial movement of the bellows (optional; blank = movement not checked; manufacturer's data)) is taken as 40 mm. (Placeholder rated axial movement; replace with the manufacturer's datasheet value.)
A44. Layout: an axial bellows absorbs the whole movement of 24.48 mm; there is no leg check and no screening. The anchors take the pressure thrust $P \cdot A_{eff}$ plus the spring force $k \cdot \Delta L$; guide friction, lateral and angular movement and the bellows' own limits are the manufacturer's. Screening only — not a piping stress analysis. (Bellows layout.)

4. Working

[9] $dT = T_{operate} - T_{install}$
$= 70\ degC - 10\ degC$
$= 60\ K$
Temperature change, operating – install

[11] $E_r = E$
$= 120\ GPa$
$= 120\ GPa$
Modulus for the reactions (E, no E_{hot} given)

[12] $D_i = OD - 2 * t_{wall}$
$= 54\ mm - 2 \times 1.5\ mm$
$= 51\ mm$
Inside diameter

[13] $A_m = pi / 4 * (OD^2 - D_i^2)$
$= \pi / 4 \times ((54\ mm)^2 - (51\ mm)^2)$
$= 247.4\ mm^2$
Metal cross-section

[14] $I_x = pi / 64 * (OD^4 - D_i^4)$
$= \pi / 64 \times ((54\ mm)^4 - (51\ mm)^4)$
$= 8.5307\ cm^4$
Second moment of area of the pipe section

[15] $Z_x = I_x / (OD / 2)$
$= 8.5307\ cm^4 / (54\ mm / 2)$
$= 3.1595\ cm^3$
Section modulus

[16] $L_{tot} = L$
$= 24\ m$
$= 24\ m$
Length that grows between the anchors

[17] $dL = \alpha * L_{tot} * abs(dT)$
$= 17\ um/(m \cdot K) \times 24\ m \times abs(60\ K)$
$= 24.48\ mm$
Thermal growth between the anchors, as a magnitude

[18] $dL_{signed} = \alpha * L_{tot} * dT$
$= 17\ um/(m \cdot K) \times 24\ m \times 60\ K$
$= 24.48\ mm$
Thermal movement with its sign: positive lengthens, negative shortens

[19] $\sigma_{res} = E * \alpha * abs(dT)$
$= 120\ GPa \times 17\ um/(m \cdot K) \times abs(60\ K)$
$= 122.4\ MPa$
Stress in the fully restrained pipe (E at the installation temperature)

[20] $F_{res} = E_r * \alpha * abs(dT) * A_m$
$= 120\ GPa \times 17\ um/(m \cdot K) \times abs(60\ K) \times 247.4\ mm^2$
$= 30.282\ kN$
Force on the anchors of the fully restrained pipe

[21] $F_{friction}$ = support friction not entered
$= \mu$ and w_{pipe} blank
$= 0\ kN$
Friction of the sliding supports (not entered: none in the anchor force)

[24] $F_{thrust} = P_b * A_{eff}$
$= 600\ kPa \times 40\ cm^2$
$= 2.4\ kN$
Bellows pressure thrust on the anchors, at the governing (design or test) pressure

[26] $F_{spring} = k_b * dL$
= 80 kN/m × 24.48 mm
= 1.9584 kN
Bellows spring force at the full movement
[27] $anchor_{force} = F_{thrust} + F_{spring} + F_{friction}$
= 2.4 kN + 1.9584 kN + 0 kN
= 4.3584 kN
Force on the anchors: pressure thrust plus spring force (plus support friction if μ and w_{pipe} are entered)
[28] $anchor_{moment}$ = axial bellows: no bending
= the movement is axial
= 0 kN·m
Moment at the anchors: none from an axial bellows
[30] $u_{bellows} = dL / dL_{rated}$
= 24.48 mm / 40 mm
= 0.612
Movement over the bellows' rated axial movement
[31] pass = the rated movement is given and met
= 1 check(s)
= 1
1 when the movement is within the bellows' rated axial movement; 0 when it is not or when no rating was entered

5. Results

SYMBOL	VALUE	DESCRIPTION
$anchor_{force}$	4.358 kN	Force on the anchors
$anchor_{moment}$	0 kN·m	Moment at the anchors
dL	24.48 mm	Thermal movement between the anchors (magnitude) (governing)
D_i	51 mm	Inside diameter
A_m	247.4 mm²	Metal cross-section
I_x	8.531 cm⁴	Second moment of area
Z_x	3.160 cm³	Section modulus
dT	60 K	Temperature change
L_{tot}	24 m	Length that grows
dL_{signed}	24.48 mm	Thermal movement, signed
σ_{res}	122.4 MPa	Fully restrained stress
F_{res}	30.28 kN	Fully restrained force
direction	"growth: the pipe lengthens; fully restrained it presses on its anchors (compression)"	Direction of the movement
$F_{friction}$	0 kN	Support friction on the run
F_{thrust}	2.400 kN	Bellows pressure thrust
F_{spring}	1.958 kN	Bellows spring force

SYMBOL	VALUE	DESCRIPTION
screen	"not applicable: the bellows absorbs the movement"	Screening result
pass	1	All checks pass
$u_{bellows}$	0.61	Movement over the rated movement

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Movement against the rated axial movement of the bellows	24.48 mm	40 mm	0.61	

7. Applicability

WARNING records: tabulated values not taken from a reference record are E, α (Material properties); OD, t_{wall} (Pipe section (outside diameter and wall)); A_{eff}, k_b, dL_{rated} (Axial bellows data). Enter each as one so its source, edition and licence travel with it. [record]
WARNING μ: the anchor force holds no support friction. Enter μ and the weight per metre, with the length on slides, to include it. [friction]

8. Verification

Template bs.pipe-expansion@1.5.1 Content hash 29443308b79dc6fc Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. ASME B31.3-2022.
2. BS EN 13480-3:2017.